

File ownership

❖ Task 1: Understanding Ownership

- Run `ls -l` in your home directory
- Identify the owner and group columns
- Check who owns your files
- Document: What is the difference between owner and group?

`Cd /home ls -l`: is used to list the directory contents '`-l`' uses a long listing format. It shows that owner has read, write, and execute permissions, group has read and execute permissions.

```
ubuntu@ip-172-31-42-193:~$ cd /home
ubuntu@ip-172-31-42-193:/home$ ls -l
total 20
drwxr-x--- 2 berlin    berlin    4096 Feb  9 06:17 berlin
drwxr-x--- 2 nairobi   nairobi   4096 Feb  9 07:11 nairobi
drwxr-x--- 2 professor professor 4096 Feb  8 11:12 professor
drwxr-x--- 2 tokyo     tokyo     4096 Feb  8 12:34 tokyo
drwxr-x--- 5 ubuntu    ubuntu    4096 Feb 10 06:46 ubuntu
ubuntu@ip-172-31-42-193:/home$ man ls
```

What is the difference between owner and group?

Ans owner: The individual user who owns the file/directory. it has full control over the file can change the permissions, delete the file/ directory.

Group: A collection of users who share access to the file/directory. it allows multiple users to collaborate with the same permissions. Users in the group get the permissions specified in the group section.

❖ Task 2: Basic chown operations

- Create file `devops-file.txt`
- Check current owner: `ls -l devops-file.txt`
- Change owner to `tokyo`
- Verify the changes.
- `Sudo chown tokyo devops-file.txt`

`touch devops_file.txt`: to create text file.

`ls -l devops_file.txt`: command is used to list directory contents '`-l`' used for long listing format.

sudo chown tokyo devops_file.txt: command is used to change file owner and group.

sudo chown berlin devops_file.txt: command is used to change file owner and group.

```
ubuntu@ip-172-31-42-193:~$ touch devops_file.txt
ubuntu@ip-172-31-42-193:~$ ls -l devops_file.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 10 07:17 devops_file.txt
ubuntu@ip-172-31-42-193:~$ chown tokyo devops_file.txt
chown: changing ownership of 'devops_file.txt': Operation not permitted
ubuntu@ip-172-31-42-193:~$ sudo chown tokyo devops_file.txt
ubuntu@ip-172-31-42-193:~$ ls -l devops_file.txt
-rw-rw-r-- 1 tokyo ubuntu 0 Feb 10 07:17 devops_file.txt
ubuntu@ip-172-31-42-193:~$ sudo chown berlin devops_file.txt
ubuntu@ip-172-31-42-193:~$ ls -l devops_file.txt
-rw-rw-r-- 1 berlin ubuntu 0 Feb 10 07:17 devops_file.txt
```

❖ Task 3: Basic chgrp operations

- Create file team_notes.txt
- Check current group: sudo groupadd heist-team
- Change file group to heist_team
- Verify the change

touch team_notes.txt: to create text file.

ls -l team_notes.txt: command is used to list directory contents '-l' used for long listing format.

sudo groupadd heist_team: command is used to create a new group.

tail -n 5 /etc/group: display output of the last part of files.

sudo chgrp heist_team team_notes.txt: is used to change group ownership.

ls -l team_notes.txt: is used to list directory contents '-l' used for long listing format.

```

ubuntu@ip-172-31-42-193:~$ touch team_notes.tx
ubuntu@ip-172-31-42-193:~$ ls -l team_notes.tx
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 10 07:39 team_notes.tx
ubuntu@ip-172-31-42-193:~$ sudo groupadd heist_team
ubuntu@ip-172-31-42-193:~$ tail -n 5 /etc/group
developers:x:1004:tokyo,berlin
admins:x:1005:berlin,professor
nairobi:x:1006:
project-team:x:1007:nairobi,tokyo
heist_team:x:1008:
ubuntu@ip-172-31-42-193:~$ sudo chgrp heist_team team_notes.tx
ubuntu@ip-172-31-42-193:~$ ls -l team_notes.tx
-rw-rw-r-- 1 ubuntu heist_team 0 Feb 10 07:39 team_notes.tx
ubuntu@ip-172-31-42-193:~$ █

```

❖ Task 4: Combined owner and group change

- Using chown you can change both owner and group together:
- Create file project_config.yaml
- Change owner to professor AND group to heist_team (one command)
- Change its owner to berlin and group to heist_team
- Syntax: sudo chown owner:group filename.

touch project_config.yaml: to create text file

sudo chown professor:heist_team project_config.yaml: is used to change file owner and group.

ls -l project_config.yaml: is used to list directory contents '-l' used for long listing format.

```

ubuntu@ip-172-31-42-193:~$ touch project_config.yaml
ubuntu@ip-172-31-42-193:~$ sudo chown professor:heist_team project_config.yaml
1
ubuntu@ip-172-31-42-193:~$ ls -l project_config.yaml
-rw-rw-r-- 1 professor heist_team 0 Feb 10 08:05 project_config.yaml
ubuntu@ip-172-31-42-193:~$ sudo chown berlin:heist_team project_config.yaml
ubuntu@ip-172-31-42-193:~$ ls -l project_config.yaml
-rw-rw-r-- 1 berlin heist_team 0 Feb 10 08:05 project_config.yaml
ubuntu@ip-172-31-42-193:~$ █

```

❖ Task 5: Recursive Ownership

➤ Create directory structure

- `mkdir -p heist-project/vault`
- `mkdir -p heist-project/plans`
- `touch heist-project/vault/gold.txt`
- `touch heist-project/plans/strategy.conf`

➤ Create group planners: `sudo groupadd planners`

➤ Change ownership of entire `heist_project/` directory:

- Owner: professor
- Group: planners
- Use recursive flag (-R)

➤ Verify all files and subdirectories changed: `ls -lR heist_project/`

`mkdir -p heist_project/vault`: it creates a directory

`mkdir -p heist_project/plans`: it creates a directory

`touch heist_project/vault/gold.txt`: it creates text file

`touch heist_project/plans/strategy.conf`: it creates text file.

`sudo groupadd planners`: it creates a new group

`sudo chown professor:planners -R heist_project/`: is used to change file owner and group.

`ls -lR heist_project/`: is used to list directory contents ‘-l’ used for long listing format.

```
ubuntu@ip-172-31-42-193: ~  
ubuntu@ip-172-31-42-193:~$ touch project_config.yaml  
ubuntu@ip-172-31-42-193:~$ sudo chown professor:heist_team project_config.yaml  
ubuntu@ip-172-31-42-193:~$ ls -l project_config.yaml  
-rw-rw-r-- 1 professor heist_team 0 Feb 10 08:05 project_config.yaml  
ubuntu@ip-172-31-42-193:~$ sudo chown berlin:heist_team project_config.yaml  
ubuntu@ip-172-31-42-193:~$ ls -l project_config.yaml  
-rw-rw-r-- 1 berlin heist_team 0 Feb 10 08:05 project_config.yaml  
ubuntu@ip-172-31-42-193:~$ mkdir -p heist_project/vault  
ubuntu@ip-172-31-42-193:~$ mkdir -p heist_project/plans  
ubuntu@ip-172-31-42-193:~$ sudo rm -rf heist_project/plans  
ubuntu@ip-172-31-42-193:~$ mkdir -p heist_project/plans  
ubuntu@ip-172-31-42-193:~$ ls -l heist_project/vault  
total 0  
ubuntu@ip-172-31-42-193:~$ ls -la heist_project/plans\  
^C  
ubuntu@ip-172-31-42-193:~$ ls -la heist_project/plans  
total 8  
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 10 08:20 .  
drwxrwxr-x 4 ubuntu ubuntu 4096 Feb 10 08:20 ..
```

```
ubuntu@ip-172-31-42-193:~$ touch heist_project/vault/gold.txt  
ubuntu@ip-172-31-42-193:~$ touch heist_project/plans/strategy.conf  
ubuntu@ip-172-31-42-193:~$ ls -la heist_project/vault/gold.txt  
ls: cannot access 'heist_project/vault/gold.txt': No such file or directory  
ubuntu@ip-172-31-42-193:~$ ls -la heist_project/vault/gold.txt  
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 10 08:22 heist_project/vault/gold.txt  
ubuntu@ip-172-31-42-193:~$ ls -la heist_project/plans/strategy.conf  
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 10 08:22 heist_project/plans/strategy.conf  
ubuntu@ip-172-31-42-193:~$ sudo groupadd planner  
ubuntu@ip-172-31-42-193:~$ sudo groupdel planner  
ubuntu@ip-172-31-42-193:~$ sudo groupadd planners  
ubuntu@ip-172-31-42-193:~$ tail -n 5 /etc/group  
tail: cannot open '/etc/group' for reading: No such file or directory  
ubuntu@ip-172-31-42-193:~$ tail -n 5 /etc/group  
admins:x:1005:berlin,professor  
nairobi:x:1006:  
project-team:x:1007:nairobi,tokyo  
heist_team:x:1008:  
planners:x:1009:  
ubuntu@ip-172-31-42-193:~$ ls -l heist_project/  
total 8  
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 10 08:22 plans  
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 10 08:22 vault
```

```
ubuntu@ip-172-31-42-193:~$ sudo chown professor:planners -R heist_project/  
ubuntu@ip-172-31-42-193:~$ ls -lR heist_project/  
heist_project/  
total 8  
drwxrwxr-x 2 professor planners 4096 Feb 10 08:22 plans  
drwxrwxr-x 2 professor planners 4096 Feb 10 08:22 vault  
  
heist_project/plans:  
total 0  
-rw-rw-r-- 1 professor planners 0 Feb 10 08:22 strategy.conf  
  
heist_project/vault:  
total 0  
-rw-rw-r-- 1 professor planners 0 Feb 10 08:22 gold.txt
```

❖ Task 6: practice challenge

- Create users: tokyo, berlin, nairobi
- Create groups: vault_team, tech_team
- Create directory: bank_heist/
- Create 3 files inside:
 - touch bank-heist/access-codes.txt
 - touch bank-heist/blueprints.pdf
 - touch bank-heist/escape-plan.txt
- Set different ownership:
 - Access_codes.txt --- owner: tokyo, group: vault_team
 - Blueprints.pdf --- owner: berlin, group: tech_team
 - Escape_plan.txt --- owner: nairobi, group: vault_team
 - Verify ls -l bank_heist/

sudo useradd -m tokyo -s /usr/bin/bash: to create a new user or default new user information.

sudo useradd -m berlin -s /usr/bin/bash: to create a new user or default new user information.

sudo useradd -m nairobi -s /usr/bin/bash: to create a new user or default new user information.

sudo groupadd vault_team: to create a new group

sudo groupadd tech_team: to create a new group

mkdir bank_heist/: to create a directory

touch bank-heist/access-codes.txt: to create a text file

touch bank-heist/blueprints.pdf: to create a pdf file

touch bank-heist/escape-plan.txt: to create a text file

sudo chown tokyo:vault_team -R bank_heist/access_codes.txt: is used to change file owner and group.

sudo chown berlin:tech_team -R bank_heist/blueprints.pdf: is used to change file owner and group.

sudo chown nairobi:vault_team -R bank_heist/escape_plan.txt: is used to change file owner and group.

```
ubuntu@ip-172-31-42-193:~$ sudo groupadd vault_team
ubuntu@ip-172-31-42-193:~$ sudo groupadd tech_team
ubuntu@ip-172-31-42-193:~$ tail -n 5 /etc/group
project-team:x:1007:nairobi,tokyo
heist_team:x:1008:
planners:x:1009:
vault_team:x:1010:
tech_team:x:1011:
ubuntu@ip-172-31-42-193:~$ mkdir -p bank_heist
ubuntu@ip-172-31-42-193:~$ touch bank_heist/access_code.txt
ubuntu@ip-172-31-42-193:~$ touch bank_heist/blueprints.pdf
ubuntu@ip-172-31-42-193:~$ touch bank_heist/escape_plan.txt
ubuntu@ip-172-31-42-193:~$ ls -l heist_project/
total 8
drwxrwxr-x 2 professor planners 4096 Feb 10 08:22 plans
drwxrwxr-x 2 professor planners 4096 Feb 10 08:22 vault
ubuntu@ip-172-31-42-193:~$ ls -l bank_heist/
total 0
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 10 09:03 access_code.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 10 09:03 blueprints.pdf
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 10 09:03 escape_plan.txt
ubuntu@ip-172-31-42-193:~$ sudo chown tokyo:vault_team
chown: missing operand after 'tokyo:vault_team'
Try 'chown --help' for more information.
ubuntu@ip-172-31-42-193:~$ sudo chown tokyo:vault_team access_code.txt
chown: cannot access 'access_code.txt': No such file or directory
ubuntu@ip-172-31-42-193:~$ sudo chown tokyo:vault_team -R bank_heist/access_c
ode.txt
ubuntu@ip-172-31-42-193:~$ sudo chown berlin:t
tape      tcpdump      tech_team  tokyo      tss        tty
ubuntu@ip-172-31-42-193:~$ sudo chown berlin:tech_team -R bank_heist/bluepri
nts.pdf
ubuntu@ip-172-31-42-193:~$ sudo chown nairobi:vault_team -R bank_heist/escape
_plan.txt
ubuntu@ip-172-31-42-193:~$ ls -lR bank_heist/
bank_heist/:
total 0
-rw-rw-r-- 1 tokyo    vault_team 0 Feb 10 09:03 access_code.txt
-rw-rw-r-- 1 berlin  tech_team 0 Feb 10 09:03 blueprints.pdf
-rw-rw-r-- 1 nairobi vault_team 0 Feb 10 09:03 escape_plan.txt
ubuntu@ip-172-31-42-193:~$
```